Geometric Sequences and Series

Concept

A geometric sequence has a constant ratio. A series adds its terms.

Term rule

$a_n = a_1 r^{n-1}$; a negative ratio alternates signs.

Finite sum

$S_n = a_1(1-r^n)/(1-r)$, $r \neq 1$; when $r=1$, $S_n = na_1$.

Infinite sum

$S_\infty = a_1/(1-r)$, only when $|r|<1$ for non-zero a_1.

Sigma notation

Count both index endpoints: $k=0$ to 3 contains four terms.

Example 1

Find term 7 and the sum of 7 terms of 5,15,45,... .

Solution

1. $r = 3$; $a_7 = 5 \times 3^6 = 3645$.

2. $S_7 = 5(3^7 - 1)/(3 - 1) = 5465$.

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Example 2

Find the infinite sum $12 + 6 + 3 + \ldots$.

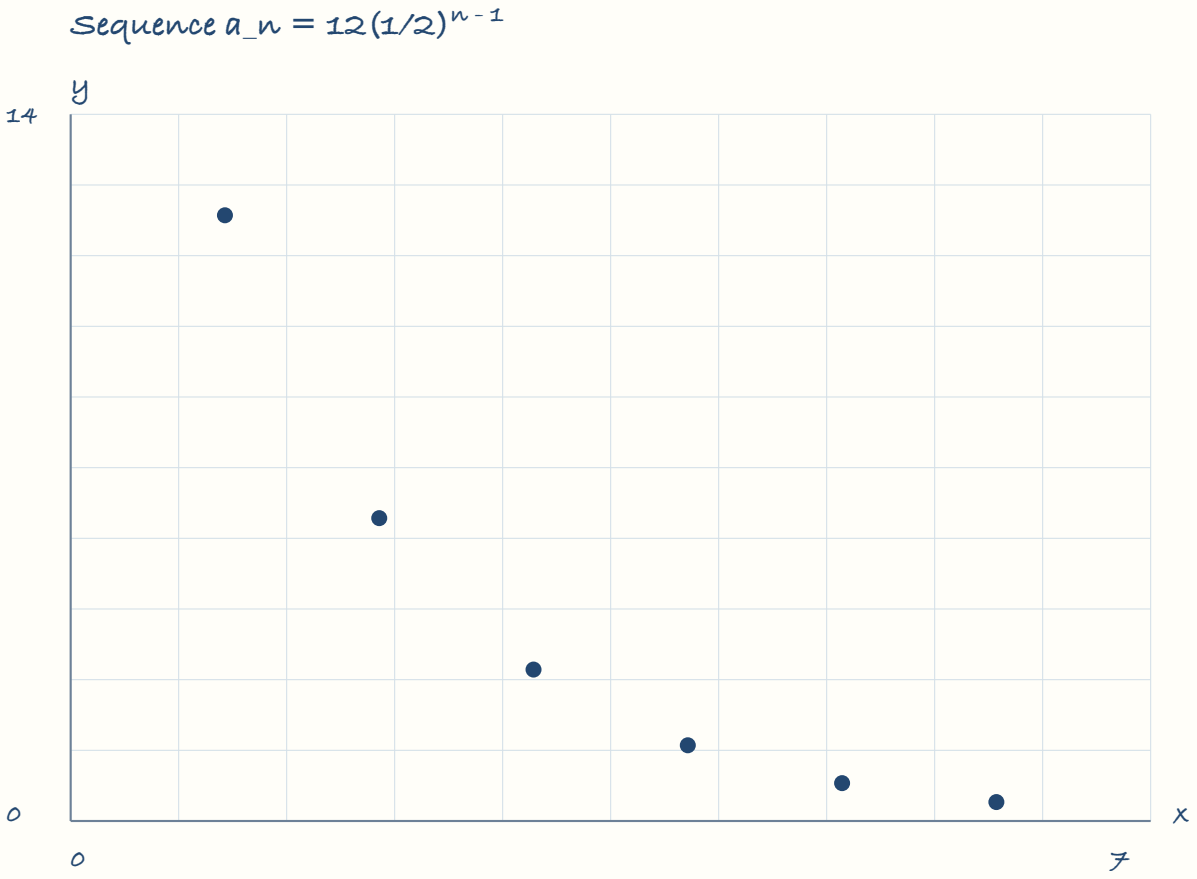
Solution

1. $r = 1/2$; $|r| < 1$, so the series converges.

2. $S_\infty = 12/(1 - 1/2) = 24$.

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Visual notes | read the graph



Term magnitudes approach zero. The sum of the infinite geometric series approaches 24.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

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Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Find term 6 of 64,32,16,… .

2. Sum the first five terms of 2,6,18,… .

3. Find the infinite sum $9 - 3 + 1 - 1/3 + \ldots$.

4. Does $4 + 6 + 9 + \ldots$ converge?

5. Expand $\sum$ from $k = 0$ to 3 of $2(5^k)$.

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Practice answers | check after trying

1. $64(1/2)^5 = 2$.

2. $2(3^5 - 1)/(3 - 1) = 242$.

3. $r = -1/3$, so $S\infty = 9/(1 + 1/3) = 27/4$.

4. No: $r = 1.5$ and terms do not approach zero.

5. $2 + 10 + 50 + 250 = 312$. There are four terms because both endpoints are included.